1. Analytical Derivation of Sphere-of-Influence Encounter Points

1.1. Problem Statement

Given two Keplerian elliptical orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI}:

$$\|\boldsymbol{A}\boldsymbol{r}_1(E_1) - \boldsymbol{B}\boldsymbol{r}_2(E_2)\| = r_{\text{SOI}} \tag{1}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

1.2. Notation

- a_i, b_i, e_i: semi-major axis, semi-minor axis, and eccentricity of orbit i (with $b_i = a_i\sqrt{1 - e_i^2}$)
- E_i: eccentric anomaly of orbit i
- $\boldsymbol{A} \in \mathbb{R}^{3\times 2}$, $\boldsymbol{B} \in \mathbb{R}^{3\times 2}$: orbital-to-ecliptic rotation matrices with components A_{jk}, B_{jk}
- r_{SOI}: radius of the planet's sphere of influence

1.3. Position in Orbital Plane

For a Keplerian ellipse, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\boldsymbol{r}_i(E_i) = \begin{pmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{pmatrix} \tag{2}$$

1.4. Distance Constraint

Square both sides of the constraint:

$$(\boldsymbol{A}\boldsymbol{r}_1 - \boldsymbol{B}\boldsymbol{r}_2)^\top (\boldsymbol{A}\boldsymbol{r}_1 - \boldsymbol{B}\boldsymbol{r}_2) = r_{\text{SOI}}^2 \tag{3}$$

Expand the left-hand side:

$$\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{r}_1 - 2\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{B}\boldsymbol{r}_2 + \boldsymbol{r}_2^\top \boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{r}_2 = r_{\text{SOI}}^2 \tag{4}$$

1.4.1. Orthonormality of $\boldsymbol{A}$ and $\boldsymbol{B}$

Since $\boldsymbol{A}$ and $\boldsymbol{B}$ are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{I}_{2\times 2}, \quad \boldsymbol{B}^\top \boldsymbol{B} = \boldsymbol{I}_{2\times 2} \tag{5}$$

Therefore the self-terms simplify:

$$\boldsymbol{r}_1^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{r}_1 = \boldsymbol{r}_1^\top \boldsymbol{r}_1 = |\boldsymbol{r}_1|^2 =: r_1^2 \tag{6}$$

$$\boldsymbol{r}_2^\top \boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{r}_2 = \boldsymbol{r}_2^\top \boldsymbol{r}_2 = |\boldsymbol{r}_2|^2 =: r_2^2 \tag{7}$$

1.4.2. The coupling matrix $\boldsymbol{C}$

Define the 2×2 matrix:

$$\boldsymbol{C} := 2\boldsymbol{A}^\top \boldsymbol{B} \tag{8}$$

This encodes the entire mutual orientation of the two orbital planes. Equation 4 becomes:

$$r_1^2 + r_2^2 - \boldsymbol{r}_1^\top \boldsymbol{C}\boldsymbol{r}_2 = r_{\text{SOI}}^2 \tag{9}$$

Note: $\boldsymbol{C}$ is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \leq 2$.

1.5. Expanding the Constraint

1.5.1. Self-terms

Using $r_i = a_i(1 - e_i \cos E_i)$:

$$r_1^2 + r_2^2 = a_1^2(1 - e_1 \cos E_1)^2 + a_2^2(1 - e_2 \cos E_2)^2 \tag{10}$$

1.5.2. Cross-term

The position vector Equation 2 factors as $\boldsymbol{r}_i = \operatorname{diag}(a_i, b_i)\boldsymbol{p}_i$, where:

$$\boldsymbol{p}_i = \begin{pmatrix} \cos E_i - e_i \\ \sin E_i \end{pmatrix} \tag{11}$$

Substituting into the cross-term:

$$\boldsymbol{r}_1^\top \boldsymbol{C} \boldsymbol{r}_2 = \boldsymbol{p}_1^\top \operatorname{diag}(a_1, b_1) \boldsymbol{C} \operatorname{diag}(a_2, b_2) \boldsymbol{p}_2 = \boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 \tag{12}$$

where the **scaled coupling matrix** is:

$$\boldsymbol{M} := \operatorname{diag}(a_1, b_1) \boldsymbol{C} \operatorname{diag}(a_2, b_2) = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} := \begin{pmatrix} C_{11} a_1 a_2 & C_{12} a_1 b_2 \\ C_{21} a_2 b_1 & C_{22} b_1 b_2 \end{pmatrix} \tag{13}$$

Expanding the bilinear form $\boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2$:

$$\begin{aligned} \boldsymbol{p}_1^\top \boldsymbol{M} \boldsymbol{p}_2 = {} & M_{11} \cos E_1 \cos E_2 - M_{11} e_2 \cos E_1 - M_{11} e_1 \cos E_2 + M_{11} e_1 e_2 \\ & + M_{12} \cos E_1 \sin E_2 - M_{12} e_1 \sin E_2 \\ & + M_{21} \sin E_1 \cos E_2 - M_{21} e_2 \sin E_1 \\ & + M_{22} \sin E_1 \sin E_2 \end{aligned} \tag{14}$$

1.5.3. Full constraint

Substituting Equation 10 and Equation 14 into Equation 9 and collecting by trigonometric structure:

$$\begin{aligned} & \underbrace{a_1^2 + a_2^2 - M_{11} e_1 e_2 - r_{\mathrm{SOI}}^2}_{\text{constant}} \\ & + \underbrace{\left(-2a_1^2 e_1 + M_{11} e_2\right)}_{\alpha_1} \cos E_1 + \underbrace{\left(-2a_2^2 e_2 + M_{11} e_1\right)}_{\alpha_2} \cos E_2 \\ & + \underbrace{M_{21} e_2}_{\beta_1} \sin E_1 + \underbrace{M_{12} e_1}_{\beta_2} \sin E_2 \\ & + a_1^2 e_1^2 \cos^2 E_1 + a_2^2 e_2^2 \cos^2 E_2 \\ & - M_{11} \cos E_1 \cos E_2 - M_{12} \cos E_1 \sin E_2 \\ & - M_{21} \sin E_1 \cos E_2 - M_{22} \sin E_1 \sin E_2 \\ & = 0 \end{aligned} \tag{15}$$

1.6. Sum-to-Product Reduction

Define sum and difference anomalies $S := E_1 + E_2$, $D := E_1 - E_2$.

1.6.1. Squared cosines

Using $\cos^2 E_i = \frac{1}{2}(1 + \cos 2E_i)$:

$$a_1^2 e_1^2 \cos^2 E_1 + a_2^2 e_2^2 \cos^2 E_2 = \frac{a_1^2 e_1^2 + a_2^2 e_2^2}{2} + \frac{a_1^2 e_1^2}{2} \cos 2E_1 + \frac{a_2^2 e_2^2}{2} \cos 2E_2 \tag{16}$$

1.6.2. Mixed products

Applying the product-to-sum identities $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$, $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$, $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$, $\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$:

$$\begin{aligned} -M_{11} \cos E_1 \cos E_2 &= -\frac{M_{11}}{2}(\cos D + \cos S) \\ -M_{22} \sin E_1 \sin E_2 &= -\frac{M_{22}}{2}(\cos D - \cos S) \\ -M_{21} \sin E_1 \cos E_2 &= -\frac{M_{21}}{2}(\sin S + \sin D) \\ -M_{12} \cos E_1 \sin E_2 &= -\frac{M_{12}}{2}(\sin S - \sin D) \end{aligned} \tag{17}$$

Collecting coefficients of $\cos D$, $\cos S$, $\sin D$, $\sin S$:

$$-\frac{M_{11}+M_{22}}{2}\cos D-\frac{M_{11}-M_{22}}{2}\cos S+\frac{M_{12}-M_{21}}{2}\sin D-\frac{M_{12}+M_{21}}{2}\sin S \tag{18}$$

1.6.3. Collected constraint

Absorbing the constant $\frac{1}{2}(a_1^2e_1^2+a_2^2e_2^2)$ into the constant term and defining:

$$K := a_1^2+a_2^2+\frac{a_1^2e_1^2+a_2^2e_2^2}{2}-M_{11}e_1e_2-r_{\mathrm{SOI}}^2 \tag{19}$$

the full constraint becomes:

$$\begin{aligned}
&\alpha_1\cos E_1+\alpha_2\cos E_2+\beta_1\sin E_1+\beta_2\sin E_2\\
&+\frac{a_1^2e_1^2}{2}\cos 2E_1+\frac{a_2^2e_2^2}{2}\cos 2E_2\\
&-\frac{M_{11}+M_{22}}{2}\cos D-\frac{M_{11}-M_{22}}{2}\cos S\\
&+\frac{M_{12}-M_{21}}{2}\sin D-\frac{M_{12}+M_{21}}{2}\sin S\\
&+K\\
&=0
\end{aligned} \tag{20}$$

where $\alpha_1=-2a_1^2e_1+M_{11}e_2$, $\alpha_2=-2a_2^2e_2+M_{11}e_1$, $\beta_1=M_{21}e_2$, $\beta_2=M_{12}e_1$.

1.7. Harmonic Addition

The terms $\alpha_1\cos E_1+\beta_1\sin E_1$ in Equation 20 have the same frequency and can be combined via the harmonic addition theorem:

$$\alpha_1\cos E_1+\beta_1\sin E_1=R_1\cos(E_1-\varphi_1) \tag{21}$$

where:

$$R_1 := \sqrt{\alpha_1^2+\beta_1^2},\quad \varphi_1 := \operatorname{atan2}(\beta_1,\alpha_1) \tag{22}$$

Similarly for E_2, D, and S:

$$\begin{aligned}
\alpha_2\cos E_2+\beta_2\sin E_2&=R_2\cos(E_2-\varphi_2)\\
-\frac{M_{11}+M_{22}}{2}\cos D+\frac{M_{12}-M_{21}}{2}\sin D&=R_D\cos(D-\varphi_D)\\
-\frac{M_{11}-M_{22}}{2}\cos S-\frac{M_{12}+M_{21}}{2}\sin S&=R_S\cos(S-\varphi_S)
\end{aligned} \tag{23}$$

where:

$$R_2 := \sqrt{\alpha_2^2+\beta_2^2},\quad \varphi_2 := \operatorname{atan2}(\beta_2,\alpha_2) \tag{24}$$

$$R_D := \frac{1}{2}\sqrt{(M_{11}+M_{22})^2+(M_{12}-M_{21})^2},\quad \varphi_D := \operatorname{atan2}(M_{12}-M_{21},-(M_{11}+M_{22})) \tag{25}$$

$$R_S := \frac{1}{2}\sqrt{(M_{11}-M_{22})^2+(M_{12}+M_{21})^2},\quad \varphi_S := \operatorname{atan2}(-(M_{12}+M_{21}),-(M_{11}-M_{22})) \tag{26}$$

Substituting back into Equation 20:

$$R_1\cos(E_1-\varphi_1)+R_2\cos(E_2-\varphi_2)+\frac{a_1^2e_1^2}{2}\cos 2E_1+\frac{a_2^2e_2^2}{2}\cos 2E_2+R_D\cos(D-\varphi_D)+R_S\cos(S-\varphi_S)+K=0 \tag{27}$$

1.8. Existence Bounds

1.8.1. Naive bound

Since each cosine lies in $[-1,1]$, the function $f(E_1,E_2)$ in Equation 27 is bounded in $[K-A,K+A]$ where:

$$A := R_1 + R_2 + \frac{a_1^2 e_1^2}{2} + \frac{a_2^2 e_2^2}{2} + R_D + R_S \tag{28}$$

A necessary condition for solutions is $|K| \leq A$. Equivalently, if $|K| > A$ then **no encounter is possible**. This treats all six cosines as independent, but they share only two degrees of freedom.

1.8.2. Separable decomposition

Split f into terms by their variable dependence:

$$f(E_1, E_2) = g_1(E_1) + g_2(E_2) + h(E_1, E_2) + K \tag{29}$$

where:

$$g_i(E_i) := R_i \cos(E_i - \varphi_i) + \frac{a_i^2 e_i^2}{2} \cos 2E_i \tag{30}$$

$$h(E_1, E_2) := R_D \cos(D - \varphi_D) + R_S \cos(S - \varphi_S) \tag{31}$$

1.8.2.1. Bounding h: amplitude depends on E_1

Since both terms of h have frequency 1 in E_2 (one via $D = E_1 - E_2$, one via $S = E_1 + E_2$), we can collect them. Expanding into $\cos E_2$ and $\sin E_2$ components:

$$h = P(E_1) \cos E_2 + Q(E_1) \sin E_2 \tag{32}$$

where:

$$\begin{aligned} P(E_1) &= R_D \cos(E_1 - \varphi_D) + R_S \cos(E_1 - \varphi_S) \\ Q(E_1) &= R_D \sin(E_1 - \varphi_D) - R_S \sin(E_1 - \varphi_S) \end{aligned} \tag{33}$$

By harmonic addition in E_2, for any fixed E_1:

$$|h| \leq \sqrt{P^2 + Q^2} =: \rho(E_1) \tag{34}$$

Computing $P^2 + Q^2$ and using $\cos A \cos B - \sin A \sin B = \cos(A + B)$:

$$\rho(E_1)^2 = R_D^2 + R_S^2 + 2R_D R_S \cos(2E_1 - \varphi_D - \varphi_S) \tag{35}$$

This varies between $(R_D - R_S)^2$ and $(R_D + R_S)^2$, so:

$$|R_D - R_S| \leq \rho(E_1) \leq R_D + R_S \tag{36}$$

The naive bound uses $\rho = R_D + R_S$ everywhere. The tighter bound recognizes that for most values of E_1, the mixed term has a smaller amplitude.

1.8.2.2. Bounding g_i: coupled harmonics

Each g_i is a sum of first and second harmonics of the same variable. Writing $\theta_i = E_i - \varphi_i$:

$$g_i = R_i \cos\theta_i + \frac{a_i^2 e_i^2}{2} \cos(2\theta_i + 2\varphi_i) \tag{37}$$

The extreme values $g_{i,\min}$ and $g_{i,\max}$ can be found by solving the 1D stationarity condition $g_i' = 0$, which is a cubic in $\cos\theta_i$ — solvable in closed form or cheaply by 1D numerical sweep. The naive bound $|g_i| \leq R_i + a_i^2 e_i^2 / 2$ is not tight because the two harmonics cannot simultaneously achieve their extremes.

1.8.3. Tighter necessary condition

Combining the exact ranges of g_i with the E_1-dependent bound on h:

$$g_{1,\min} + g_{2,\min} - (R_D + R_S) + K \leq f \leq g_{1,\max} + g_{2,\max} + (R_D + R_S) + K \tag{38}$$

No encounter is possible if the interval does not contain zero:

$$g_{1,\max} + g_{2,\max} + (R_D + R_S) + K < 0 \quad \text{or} \quad g_{1,\min} + g_{2,\min} - (R_D + R_S) + K > 0 \tag{39}$$

For an even tighter check, one can sweep E_1 and use the E_1-dependent amplitude $\rho(E_1)$ from Equation 35 together with the exact $g_1(E_1)$:

$$\min_{E_2} f = g_1(E_1) + g_{2,\,\min} - \rho(E_1) + K \tag{40}$$

$$\max_{E_2} f = g_1(E_1) + g_{2,\,\max} + \rho(E_1) + K \tag{41}$$

If $g_1(E_1) + g_{2,\,\max} + \rho(E_1) + K < 0$ for all E_1, no encounter exists. This reduces the 2D problem to a 1D sweep.